

10.1 Clustering and K-Means

Partition points into spherical clusters.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

Figures and notes follow Géron, *Hands-On Machine Learning*; Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*; and Theodoridis.

1. Central idea

Clustering groups similar instances. No labels.

- **Customer segmentation.** Split customers by behavior, purchases, or how they interact. Different groups, different products and messages.
 - **Data analysis.** Find the clumps first. Then look at each clump, not at every row at once.
 - **Dimensionality reduction.** Represent a cloud by its clusters instead of by every point. Useful when you want a picture of high-dimensional data.
 - **Anomaly detection.** A point that fits no cluster is a candidate outlier: a defect, a fraud, odd behavior.
 - **Semi-supervised learning.** A few labels, then copy each label onto the rest of its cluster.
 - **Image segmentation.** Cluster pixels by color. Replace each pixel by the mean color of its cluster; fewer colors, easier objects to track.
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2. k -means

Lloyd proposed it at Bell Labs in 1957 for pulse-code modulation; it left the company in 1982 as “Least square quantization in PCM.” Fast. Often a few iterations.

Write $C_1, \dots, C_K$ for the index sets of the clusters. They **partition** the N observations:

1. $C_1 \cup C_2 \cup \dots \cup C_K = \{1, \dots, N\}$ — every row sits in some cluster.
2. $C_i \cap C_j = \emptyset$ for $i \neq j$ — clusters do not overlap.

A good clustering makes **within-cluster variation** small. Solve

$$\min_{C_1, C_2, \dots, C_K} \sum_{k=1}^K W(C_k).$$

Partition so the total within-cluster variation, summed over all K clusters, is as small as possible.

A typical $W(C_k)$ is squared Euclidean distance:

$$W(C_k) = \frac{1}{|C_k|} \sum_{i, i' \in C_k} \sum_{j=1}^P (x_{ij} - x_{i'j})^2,$$

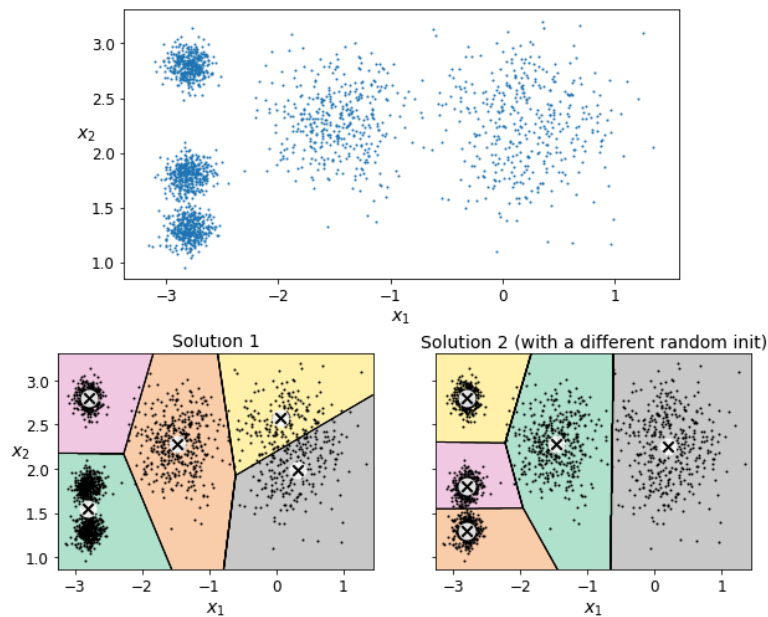
where $|C_k|$ is the number of observations in cluster k . All pairwise squared distances inside the cluster, divided by the cluster size.

3. Algorithm

1. **Initialization.** Pick K centroids. Random data points, or k -means++ (below).
 2. **Assignment.** Each point goes to the nearest centroid. Euclidean distance is the usual metric; others are allowed. That step is the partition.
 3. **Update.** Each centroid becomes the **mean** of the points assigned to it.
 4. **Convergence.** Stop when centroids (or assignments) barely move.
 5. Repeat steps 2–4. Each pass tries to cut total within-cluster variation.
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4. Why two runs disagree

Random starts. Different initial centroids, different assignments and final centers.



The objective has **local optima**. A new start can land in a different hole. The stopping rule can also fire on a slightly different iterate.

k -means++ (Arthur and Vassilvitskii, 2006) is a better start than uniform draws:

- Draw c_1 uniformly from the data.
- Draw the next c_i as instance x_i with probability

$$\frac{D(x_i)^2}{\sum_{j=1}^m D(x_j)^2},$$

where $D(x_i)$ is the distance from x_i to the nearest centroid already chosen. Far points are more likely. - Repeat until you have K centroids.

Then ordinary k -means. Suboptimal solutions become rarer, so you can cut `n_init`. In scikit-learn, `init="k-means++"` is the default.

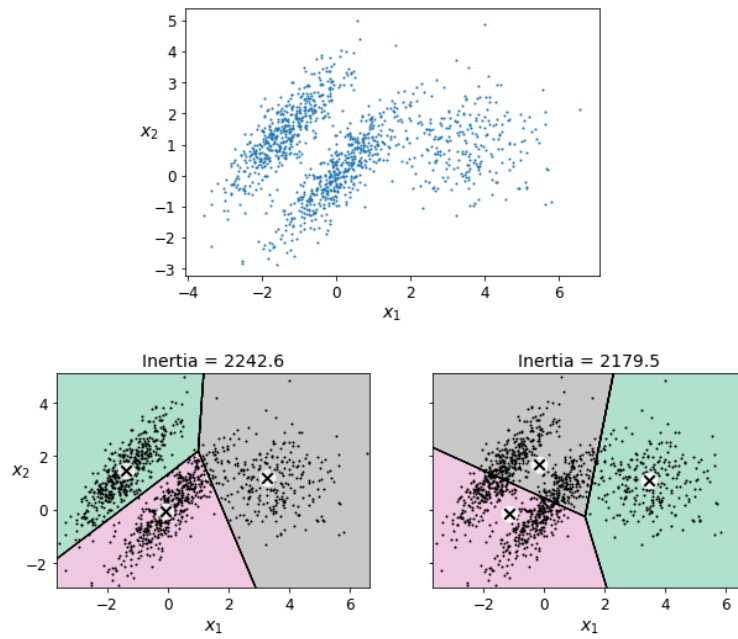
Elkan. Skip many distance computations with the triangle inequality ($AC \leq AB + BC$) and with lower and upper bounds on distances to centroids (Elkan, 2003). Set `algorithm="elkan"`. Dense data only. Sparse data uses the full algorithm.

5. Evaluation

Tool	What it tells you
Inertia	Within-cluster sum of squares. Lower is tighter.
Silhouette score	Cohesion inside a cluster versus separation from others. Range -1 to 1 ; higher is better.
Elbow method	Plot inertia against K . The “elbow” is where the drop flattens — a candidate for K .
Adjusted Rand index (ARI)	Agreement with another clustering. Range -1 to 1 .
Visual inspection	Scatter plots (or similar). Do the clumps look real?
Cross-validation	Holdout or k -fold, if you have a way to score unseen rows.

6. Limits

k -means is a poor fit when clusters have **varying sizes**, **different densities**, or **non-spherical** shapes.



Practice

1. k -means assumes spherical clusters. Give a data set where that is wrong.
2. What do you do when k is not given?
3. Why can two runs on the same matrix return different partitions?